

Integer Polynomials

The ring $k[X]$ of polynomials in one variable X over a field k forms a Euclidean domain. What can we say about division in the ring $\mathbb{Z}[X]$ of polynomials with coefficients in the ring $\mathbb{Z}$ of integers?

The classical theorem that $\sqrt{2}$ is irrational can be seen as the statement that the polynomial $X^2 - 2$ is *irreducible* as an element of $\mathbb{Q}[X]$. Since $X^2 = 2$ has no *integer* solutions with n , one can show that $X^2 - 2$ is irreducible as an element of $\mathbb{Z}[X]$. Can we make similar assertions about other polynomials?

Content of a polynomial

A general approach to such problems is due to Gauss and goes as follows.

Given a polynomial $P \in \mathbb{Z}[X]$ with coefficients in the ring of integers, we define the *content* $c(P)$ to be the greatest common divisor of the coefficients of the coefficients of P .

Suppose a prime p divides all the coefficients of the product PQ of two elements of $\mathbb{Z}[X]$. Writing $P = \sum_{k=0}^d a_k X^k$ and $Q = \sum_{k=0}^e b_k X^k$ we note that

$$PQ = \sum_{k=0}^{d+e} c_k X^k \text{ where } c_k = \sum_{m=0}^k a_m b_{k-m}$$

where we take a_m (respectively, b_k) to be 0 outside the range $[0, d]$ (respectively, $[0, e]$). We are given that p divides c_k for all k . Suppose that p does *not* divide b_k for some k and assume that k is the smallest such element of $[0, e]$. It follows that p divides *every* term in the expression $c_k - \sum_{m=1}^k a_m b_{k-m}$. Hence p divides the value $a_0 b_k$ of this expression. Since p does not divide b_k , it follows that p divides a_0 .

More generally, in the expression $c_{k+r} - \sum_{m=0}^{r-1} a_m b_{k+r-m} - \sum_{m=r+1}^{k+r} a_m b_{k+r-m}$, we see that p divides b_{k+r-m} for m in $[r+1, k+r]$ and (by induction on r) p divides a_m for m in $[0, r-1]$. It follows that p divides the value $a_r b_k$ of this expression. Since p does not divide b_k , it follows that p divides a_r .

By induction, we see that p divides *every* coefficient of P . We conclude that:

If a prime p divides the content $c(PQ)$ of a product of two integer polynomials, then either p divides $c(P)$ or it divides $c(Q)$.

Using the unique factorisation theorem for integers, we conclude the equation $c(PQ) = c(P)c(Q)$. In other words:

The content $c(PQ)$ of the product of two integer polynomials equals the product $c(P)c(Q)$ of their contents.

Clearing denominators and monic polynomials

Given a polynomial P in $\mathbb{Z}[X]$, it is clear that $P' = P/c(P)$ is also a polynomial in $\mathbb{Z}[X]$. Moreover, we note that $c(P') = 1$.

Given any polynomial P in $\mathbb{Q}[X]$, we can “collect” the denominators of the coefficients of P to find a number n such that $n \cdot P$ lies in $\mathbb{Z}[X]$.

Note that the polynomial $P' = (n \cdot P)/c(n \cdot P)$ is in $\mathbb{Z}[X]$ and satisfies $c(P') = 1$.

Given any polynomial P in $\mathbb{Q}[X]$, there is a non-zero rational number r such that rP is in $\mathbb{Z}[X]$ and $c(rP) = 1$.

Among polynomials P in $\mathbb{Z}[X]$ with $c(P) = 1$, a special role is played by polynomials whose *leading* coefficient is 1.

A polynomial P in $\mathbb{Z}[X]$ is said to be *monic* if it has leading coefficient 1.

Note that a monic polynomial P has content 1.

Factorisation of polynomials in $\mathbb{Q}[X]$ and $\mathbb{Z}[X]$

Given a polynomial P in $\mathbb{Q}[X]$, suppose that it has a factorisation $P = QR$ with Q and R as polynomials in $\mathbb{Q}[X]$.

We have a non-zero rational number r such that rP is in $\mathbb{Z}[X]$ and $c(rP) = 1$. Moreover, we see that $rP = Q(rR)$ is a factorisation of rP .

Now, let s and t be non-zero rational numbers such that sQ and trR are in $\mathbb{Z}[X]$ with $c(sQ) = 1 = c(trR)$. We see that $strP = (sQ)(trR)$ is factorisation in $\mathbb{Z}[X]$; moreover, $c(strP) = c(sQ)c(trR) = 1$.

Given a factorisation $P = QR$ in $\mathbb{Q}[X]$ there are non-zero rational numbers a, b, c such that aP, bQ and cR are in $\mathbb{Z}[X]$ with $c(aP) = 1 = c(bQ) = c(cR)$ such that $aP = (bQ)(cR)$ is a factorisation in $\mathbb{Z}[X]$.

In other words, the problem of factorisation of polynomials in $\mathbb{Q}[X]$ can *always* be reduced to a problem of factorisation of polynomials with content 1 in $\mathbb{Z}[X]$.

One useful observation is that if $P = QR$ in $\mathbb{Z}[X]$ then the leading coefficient of P is the product of the leading coefficients of Q and R . In particular, if P has leading coefficient 1, then, by changing signs of both Q and R if necessary, we may conclude that Q and R are monic as well.

If P is a monic polynomial in $\mathbb{Z}[X]$ and it can be factorised in $\mathbb{Q}[X]$, then there are monic polynomials Q and R in $\mathbb{Z}[X]$ such that $P = QR$.

If $X^2 - n$ has a rational solution then it has a factorisation of the form $(X - a)(X - b)$. By the above statement, when n is an integer, we can take a and b to be integers. This means that $a^2 = n$. So if an integer n is the square of a

rational number, then it is the square of an integer. Note that this also means that $X^2 - n$ is irreducible in $\mathbb{Q}[X]$ when n is not the square of an integer.

Similarly, if $X^k - n$ has a root in rationals, then it has a factorisation of the form $(X - a)(X^{k-1} + P(X))$ where $P(X)$ is a polynomial of degree $k - 2$. By the above statement, when n is an integer, we can take a and the coefficients of P to be integers. This means that $a^k = n$. So if an integer n is the k -th power of a rational number, then it is the k -th power of an integer.

As in the case of $k = 2$, we can use this to show that when n is not an integer cube, the polynomial that $X^3 - n$ is irreducible in $\mathbb{Q}[X]$.

However, this *does not* show that $X^k - n$ is irreducible when n is not an integer k -th power when $k \geq 4$. In fact, $X^4 - 4 = (X^2 - 2)(X^2 + 2)$. Thus we need a more sophisticated approach to prove the irreducibility of polynomials of degree at least 4.

Algorithmic test for irreducibility

Given a polynomial $P(X)$ with coefficients in $\mathbb{Q}$, we now provide an algorithm to check for its irreducibility. Note that this algorithm is *not* the most efficient. The more efficient algorithm is due to Lenstra-Lenstra-Lovacs but it is quite a bit more sophisticated. (More details can be found on the Wikipedia page on the factorisation of polynomials.)

The basic idea behind this test is that a polynomial of degree d in one variable over a domain R of characteristic 0 is completely determined by its values at $d + 1$ *distinct* elements of R .

Using the results of an earlier section, we can replace $P(X)$ by a rational multiple of it with coefficients in $\mathbb{Z}$ whose content $c(P)$ is 1. We will assume that we have done so.

To start with we consider the values $a_0 = P(0), \dots, a_d = P(d)$ of the polynomial $P(X)$ of degree d . (As mentioned above, we can take any other $d + 1$ integers in place of $0, \dots, d$.)

Using the results of an earlier section, if $P(X)$ has a factor then it has a factor $Q(X)$ with integer coefficients which are integers. Moreover, $P(X)/Q(X) = R(X)$ *also* has integer coefficients. It follows that $a_i = P(i) = Q(i)R(i)$ where $Q(i)$ and $R(i)$ are integers. If $Q(X)$ has degree e where $0 < e < d$, then Q is determined by its values $Q(0), \dots, Q(e)$ which are factors of $a_0, \dots, a_e$ respectively.

For each choice of integer factors $b_i | a_i$ for $i = 0, \dots, e$, we can write the *unique* polynomial $Q(X)$ such that $Q(i) = b_i$ for $i = 0, \dots, e$ using Lagrange interpolation:

$$Q(X) = \sum_{m=0}^e b_m \cdot \frac{\prod_{k=0; k \neq m}^e (X - k)}{\prod_{k=0; k \neq m}^e (m - k)}$$

Note that, *a priori*, this polynomial has coefficients in $\mathbb{Q}$. We can then check whether it is a factor of $P(X)$, by long division in $\mathbb{Q}[X]$.

Exercise: Use the idea above to check that $X^3 + 7X + 7$ is irreducible in $\mathbb{Q}[X]$.